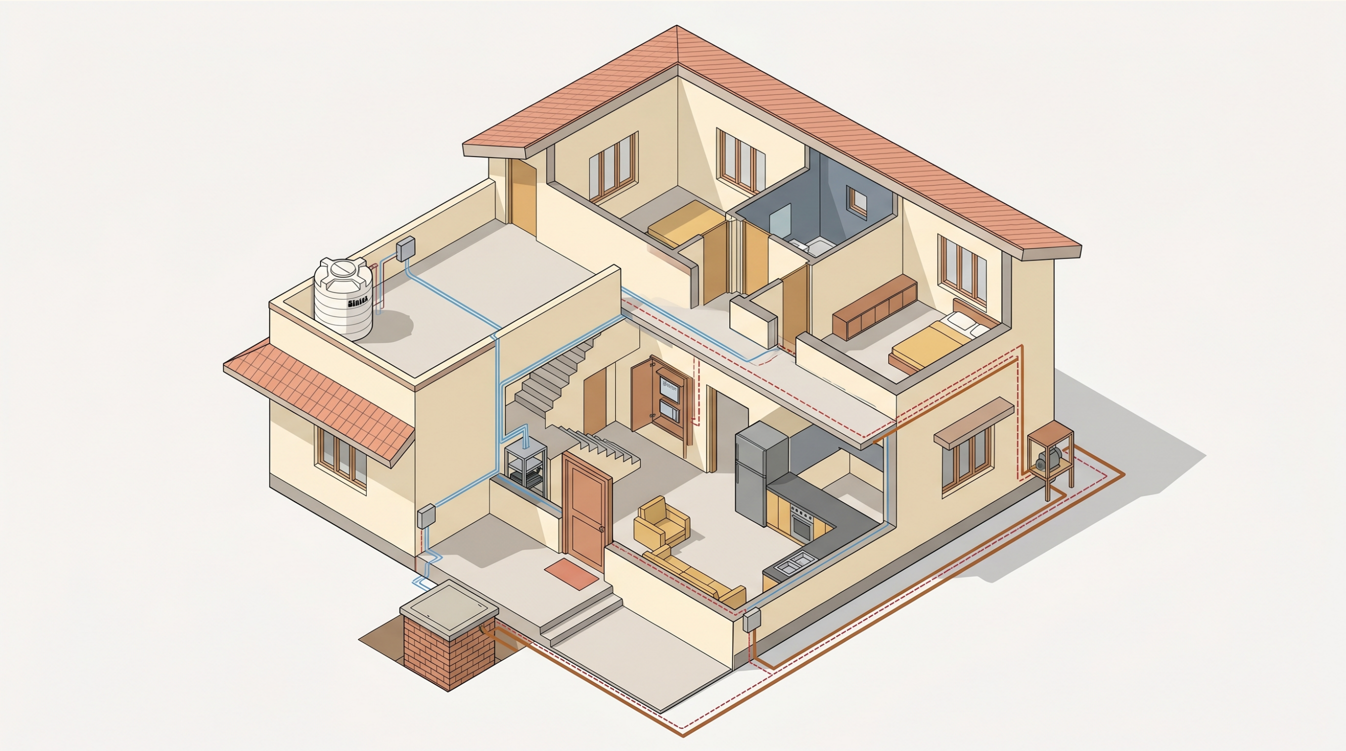
Water Automation
Visual Handoff

Three pictures + one floor plan. That is it.



AT A GLANCE

Sintex tank on the terrace. Sump in the porch. Server in the staircase. Starters + Sonoff inside the foyer cupboard. Sensor data flows over Cat6 / PoE. Float wires are the hardware failsafe. Motor power runs from the cupboard out.

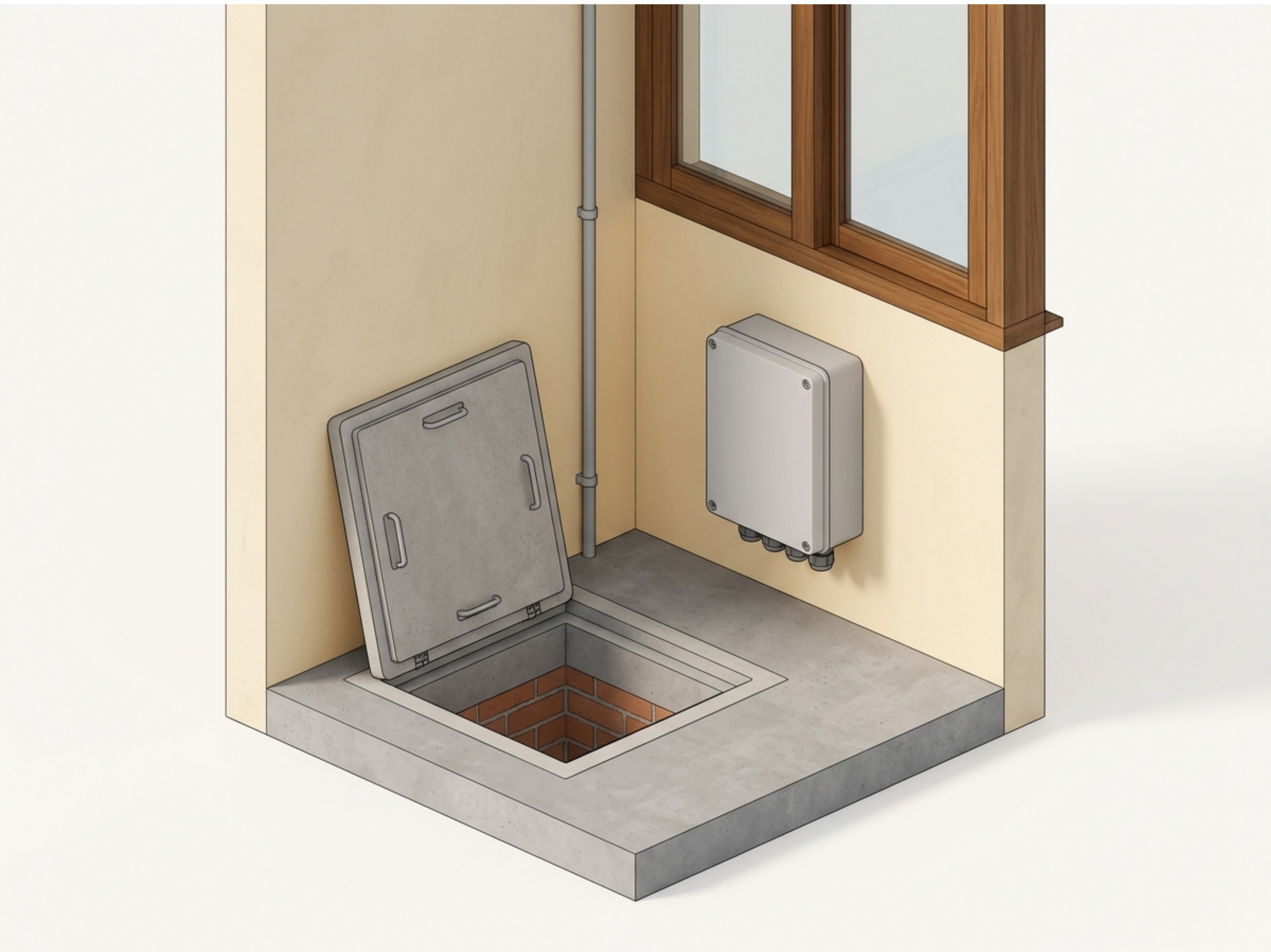
HAND THIS TO

Electrician (pre-plaster) / Plumber / Owner (reference)

02

Where the Sump JB goes

Porch West wall, 300 mm above floor, beside the manhole

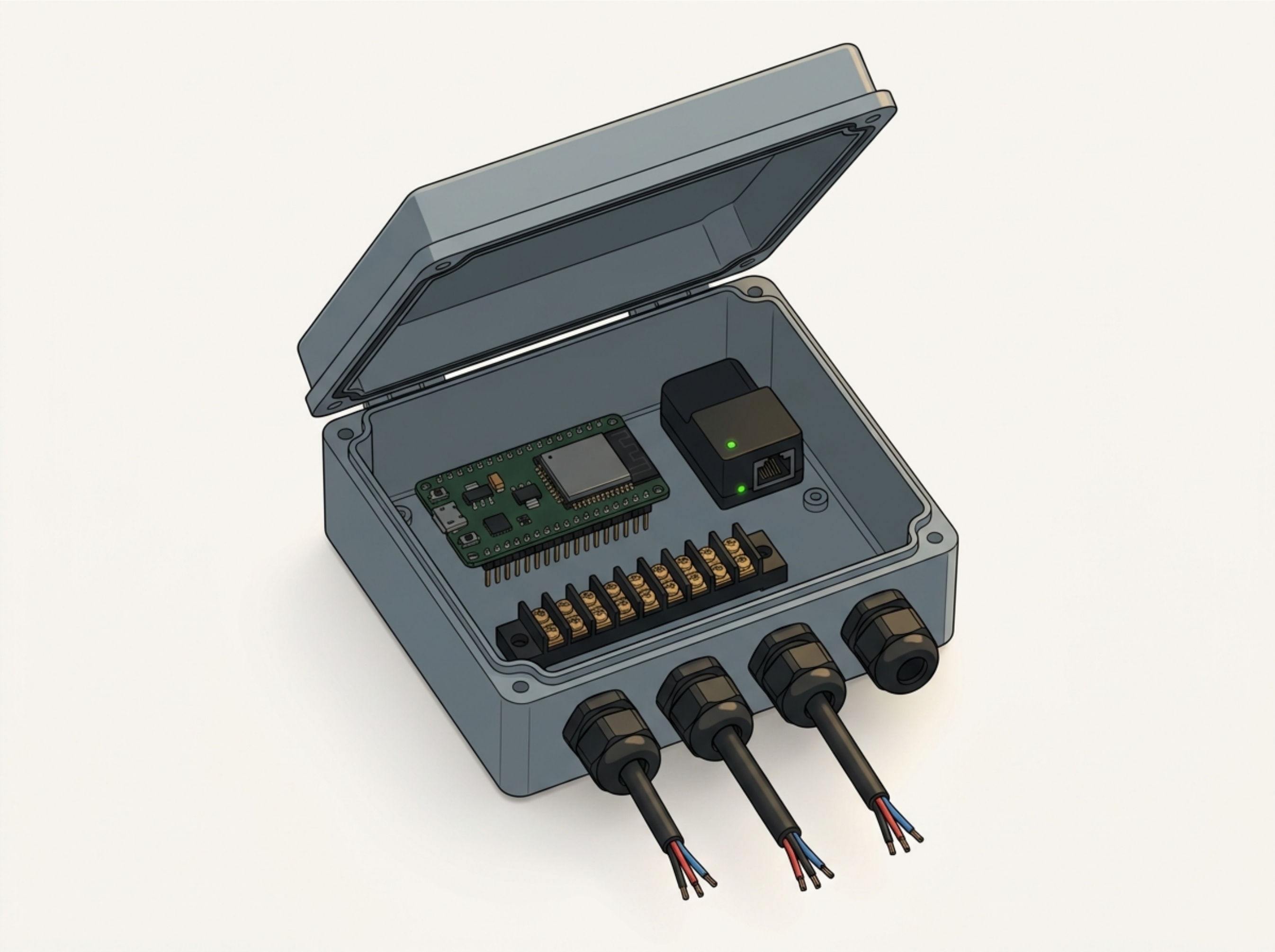


WHAT THIS PICTURE SHOWS

Junction box (grey) mounted at knee-height on the porch west wall, BESIDE the manhole (not directly above). The manhole cover, when lifted open, hinges on its west edge and lies flat against this same wall - so the JB must sit to one side. Window sill above keeps the box well clear of the glass. Conduit drops down the wall surface to the sump.

What goes inside the JB

ESP32 + PoE splitter + terminal block + 4 cable glands



WHAT THIS PICTURE SHOWS

Box lid lifted. Inside: small ESP32 board (top-left) reads the sensor and talks to the server. PoE splitter (top-right) takes 48 V off the Cat6 cable and converts to 5 V to power the board. Terminal block (bottom) is where wires land. 4 cable glands at the bottom face - the cables always enter from below so water sheds away.

04

The 7 conduits

What to lay before plastering

Conduit ID	Size	From -> To	Carries	
C-Sintex-1	20 mm grey	Server -> Sintex JB (terrace)	Cat6 (data + PoE)	
C-Sintex-2	16 mm	Sintex JB -> DB cupboard	Float wire (220 V)	
C-Sump-1	20 mm grey	Server -> Sump JB (porch W wall)	Cat6 (data + PoE)	
C-Sump-2	16 mm	Sump JB -> DB cupboard	Float wire (220 V)	
C-DB-Backup	20 mm grey	Server -> DB cupboard	Empty (pull string only)	
C-Motor-P1	25 mm blue	DB cupboard -> Borewell head	Motor power (4 sqmm arm.)	
C-Motor-P2	25 mm blue	DB cupboard -> P2 cage (E wall)	Motor power (2.5 sqmm)	

RULES THAT DON'T BEND

(1) Data conduits and mains conduits NEVER share. Run them parallel, 50 mm apart. (2) Pull string in every conduit before plastering. (3) Both sensor JBs are IP66 polycarbonate, 250 x 200 x 120 mm preferred. (4) Cable glands always on the bottom face. (5) Sump JB: porch W wall, 300 mm above floor, beside the manhole.

FULL DETAIL

See WATER_AUTOMATION.pdf for the long version - same content as this with per-conduit routing, materials checklist, sequencing, and an electrician acceptance checklist. This 4-page Visual Handoff is the short-form companion.